

Statistical Significance Testing Results

Bonferroni corrected  $\alpha = 0.0004$   
\*p<0.05, \*\*p<0.01, \*\*\*p<0.001

| Comparison                                                            | Mean Difference | p-value | Significance | Bonferroni Significant | Cohen's d | Effect Size |
|-----------------------------------------------------------------------|-----------------|---------|--------------|------------------------|-----------|-------------|
| inception3d vs LeViT-RL                                               | 0.0849          | 0.0062  | **           | No                     | 2.362     | Large       |
| inception3d vs LeVit-Attention                                        | 0.0325          | 0.4001  |              | No                     | 0.421     | Small       |
| inception3d vs 3dcnn                                                  | 0.0147          | 0.4784  |              | No                     | 0.349     | Small       |
| inception3d vs cnnlstm                                                | -0.0767         | 0.0049  | **           | No                     | -2.519    | Large       |
| inception3d vs attention_pool_nolnitWeights                           | -0.2714         | 0.0000  | ***          | Yes                    | -9.841    | Large       |
| inception3d vs mean_pool_extra3                                       | -0.3170         | 0.0001  | ***          | Yes                    | -6.993    | Large       |
| inception3d vs singletask_extra3                                      | -0.3179         | 0.0002  | ***          | Yes                    | -6.170    | Large       |
| inception3d vs uniform_extra3                                         | -0.3087         | 0.0001  | ***          | Yes                    | -6.873    | Large       |
| inception3d vs no_rl_full_train                                       | -0.2695         | 0.0001  | ***          | Yes                    | -7.542    | Large       |
| inception3d vs original_nolnitWeights                                 | -0.2758         | 0.0002  | ***          | Yes                    | -6.120    | Large       |
| inception3d vs original                                               | -0.2814         | 0.0000  | ***          | Yes                    | -8.443    | Large       |
| inception3d vs vivit                                                  | -0.1798         | 0.0001  | ***          | Yes                    | -6.294    | Large       |
| inception3d vs attention_pool_extra3_full_train                       | -0.3261         | 0.0001  | ***          | Yes                    | -7.929    | Large       |
| inception3d vs attention_pool_extra4_k1                               | -0.3065         | 0.0005  | ***          | No                     | -4.634    | Large       |
| inception3d vs attention_pool_extra4_k8                               | -0.3031         | 0.0004  | ***          | No                     | -4.788    | Large       |
| LeViT-RL vs LeVit-Attention                                           | -0.0524         | 0.0794  |              | No                     | -1.046    | Large       |
| LeViT-RL vs 3dcnn                                                     | -0.0702         | 0.0239  | *            | No                     | -1.585    | Large       |
| LeViT-RL vs cnnlstm                                                   | -0.1615         | 0.0034  | **           | No                     | -2.791    | Large       |
| LeViT-RL vs attention_pool_nolnitWeights                              | -0.3563         | 0.0000  | ***          | Yes                    | -14.868   | Large       |
| LeViT-RL vs mean_pool_extra3                                          | -0.4019         | 0.0000  | ***          | Yes                    | -10.924   | Large       |
| LeViT-RL vs singletask_extra3                                         | -0.4028         | 0.0000  | ***          | Yes                    | -13.776   | Large       |
| LeViT-RL vs uniform_extra3                                            | -0.3935         | 0.0000  | ***          | Yes                    | -12.838   | Large       |
| LeViT-RL vs no_rl_full_train                                          | -0.3544         | 0.0000  | ***          | Yes                    | -10.963   | Large       |
| LeViT-RL vs original_nolnitWeights                                    | -0.3607         | 0.0000  | ***          | Yes                    | -11.860   | Large       |
| LeViT-RL vs original                                                  | -0.3663         | 0.0000  | ***          | Yes                    | -15.559   | Large       |
| LeViT-RL vs vivit                                                     | -0.2647         | 0.0000  | ***          | Yes                    | -14.279   | Large       |
| LeViT-RL vs attention_pool_extra3_full_train                          | -0.4110         | 0.0000  | ***          | Yes                    | -13.927   | Large       |
| LeViT-RL vs attention_pool_extra4_k1                                  | -0.3913         | 0.0001  | ***          | Yes                    | -7.680    | Large       |
| LeViT-RL vs attention_pool_extra4_k8                                  | -0.3880         | 0.0001  | ***          | Yes                    | -7.830    | Large       |
| LeVit-Attention vs 3dcnn                                              | -0.0178         | 0.4899  |              | No                     | -0.340    | Small       |
| LeVit-Attention vs cnnlstm                                            | -0.1092         | 0.0489  | *            | No                     | -1.252    | Large       |
| LeVit-Attention vs attention_pool_nolnitWeights                       | -0.3039         | 0.0005  | ***          | No                     | -4.636    | Large       |
| LeVit-Attention vs mean_pool_extra3                                   | -0.3495         | 0.0001  | ***          | Yes                    | -7.293    | Large       |
| LeVit-Attention vs singletask_extra3                                  | -0.3504         | 0.0000  | ***          | Yes                    | -9.194    | Large       |
| LeVit-Attention vs uniform_extra3                                     | -0.3412         | 0.0001  | ***          | Yes                    | -7.643    | Large       |
| LeVit-Attention vs no_rl_full_train                                   | -0.3020         | 0.0007  | ***          | No                     | -4.221    | Large       |
| LeVit-Attention vs original_nolnitWeights                             | -0.3083         | 0.0000  | ***          | Yes                    | -8.596    | Large       |
| LeVit-Attention vs original                                           | -0.3139         | 0.0003  | ***          | Yes                    | -5.369    | Large       |
| LeVit-Attention vs vivit                                              | -0.2123         | 0.0007  | ***          | No                     | -4.191    | Large       |
| LeVit-Attention vs attention_pool_extra3_full_train                   | -0.3587         | 0.0001  | ***          | Yes                    | -7.890    | Large       |
| LeVit-Attention vs attention_pool_extra4_k1                           | -0.3390         | 0.0001  | ***          | Yes                    | -6.570    | Large       |
| LeVit-Attention vs attention_pool_extra4_k8                           | -0.3357         | 0.0001  | ***          | Yes                    | -6.579    | Large       |
| 3dcnn vs cnnlstm                                                      | -0.0913         | 0.0135  | *            | No                     | -1.888    | Large       |
| 3dcnn vs attention_pool_nolnitWeights                                 | -0.2861         | 0.0002  | ***          | Yes                    | -5.641    | Large       |
| 3dcnn vs mean_pool_extra3                                             | -0.3317         | 0.0000  | ***          | Yes                    | -10.589   | Large       |
| 3dcnn vs singletask_extra3                                            | -0.3326         | 0.0001  | ***          | Yes                    | -7.241    | Large       |
| 3dcnn vs uniform_extra3                                               | -0.3233         | 0.0000  | ***          | Yes                    | -8.385    | Large       |
| 3dcnn vs no_rl_full_train                                             | -0.2842         | 0.0005  | ***          | No                     | -4.733    | Large       |
| 3dcnn vs original_nolnitWeights                                       | -0.2905         | 0.0000  | ***          | Yes                    | -11.069   | Large       |
| 3dcnn vs original                                                     | -0.2961         | 0.0001  | ***          | Yes                    | -6.464    | Large       |
| 3dcnn vs vivit                                                        | -0.1945         | 0.0001  | ***          | Yes                    | -6.344    | Large       |
| 3dcnn vs attention_pool_extra3_full_train                             | -0.3408         | 0.0000  | ***          | Yes                    | -10.526   | Large       |
| 3dcnn vs attention_pool_extra4_k1                                     | -0.3211         | 0.0003  | ***          | Yes                    | -5.327    | Large       |
| 3dcnn vs attention_pool_extra4_k8                                     | -0.3178         | 0.0002  | ***          | Yes                    | -5.747    | Large       |
| cnnlstm vs attention_pool_nolnitWeights                               | -0.1947         | 0.0005  | ***          | No                     | -4.643    | Large       |
| cnnlstm vs mean_pool_extra3                                           | -0.2403         | 0.0005  | ***          | No                     | -4.589    | Large       |
| cnnlstm vs singletask_extra3                                          | -0.2413         | 0.0008  | ***          | No                     | -4.059    | Large       |
| cnnlstm vs uniform_extra3                                             | -0.2320         | 0.0005  | ***          | No                     | -4.502    | Large       |
| cnnlstm vs no_rl_full_train                                           | -0.1929         | 0.0007  | ***          | No                     | -4.191    | Large       |
| cnnlstm vs original_nolnitWeights                                     | -0.1992         | 0.0011  | **           | No                     | -3.758    | Large       |
| cnnlstm vs original                                                   | -0.2048         | 0.0006  | ***          | No                     | -4.443    | Large       |
| cnnlstm vs vivit                                                      | -0.1031         | 0.0065  | **           | No                     | -2.325    | Large       |
| cnnlstm vs attention_pool_extra3_full_train                           | -0.2495         | 0.0003  | ***          | Yes                    | -5.055    | Large       |
| cnnlstm vs attention_pool_extra4_k1                                   | -0.2298         | 0.0014  | **           | No                     | -3.507    | Large       |
| cnnlstm vs attention_pool_extra4_k8                                   | -0.2265         | 0.0014  | **           | No                     | -3.558    | Large       |
| attention_pool_nolnitWeights vs mean_pool_extra3                      | -0.0456         | 0.0628  |              | No                     | -1.144    | Large       |
| attention_pool_nolnitWeights vs singletask_extra3                     | -0.0465         | 0.0326  | *            | No                     | -1.435    | Large       |
| attention_pool_nolnitWeights vs uniform_extra3                        | -0.0373         | 0.0541  |              | No                     | -1.207    | Large       |
| attention_pool_nolnitWeights vs no_rl_full_train                      | 0.0019          | 0.7245  |              | No                     | 0.169     | Negligible  |
| attention_pool_nolnitWeights vs original_nolnitWeights                | -0.0044         | 0.8063  |              | No                     | -0.117    | Negligible  |
| attention_pool_nolnitWeights vs original                              | -0.0100         | 0.2162  |              | No                     | -0.656    | Medium      |
| attention_pool_nolnitWeights vs vivit                                 | 0.0916          | 0.0008  | ***          | No                     | 4.026     | Large       |
| attention_pool_nolnitWeights vs attention_pool_extra3_full_train      | -0.0548         | 0.0182  | *            | No                     | -1.724    | Large       |
| attention_pool_nolnitWeights vs attention_pool_extra4_k1              | -0.0351         | 0.1655  |              | No                     | -0.758    | Medium      |
| attention_pool_nolnitWeights vs attention_pool_extra4_k8              | -0.0317         | 0.1999  |              | No                     | -0.686    | Medium      |
| mean_pool_extra3 vs singletask_extra3                                 | -0.0009         | 0.9417  |              | No                     | -0.035    | Negligible  |
| mean_pool_extra3 vs uniform_extra3                                    | 0.0083          | 0.2837  |              | No                     | 0.553     | Medium      |
| mean_pool_extra3 vs no_rl_full_train                                  | 0.0475          | 0.0775  |              | No                     | 1.056     | Large       |
| mean_pool_extra3 vs original_nolnitWeights                            | 0.0412          | 0.0066  | **           | No                     | 2.320     | Large       |
| mean_pool_extra3 vs original                                          | 0.0356          | 0.0416  | *            | No                     | 1.324     | Large       |
| mean_pool_extra3 vs vivit                                             | 0.1372          | 0.0002  | ***          | Yes                    | 6.226     | Large       |
| mean_pool_extra3 vs attention_pool_extra3_full_train                  | -0.0092         | 0.1006  |              | No                     | -0.951    | Large       |
| mean_pool_extra3 vs attention_pool_extra4_k1                          | 0.0105          | 0.5437  |              | No                     | 0.296     | Small       |
| mean_pool_extra3 vs attention_pool_extra4_k8                          | 0.0138          | 0.3336  |              | No                     | 0.491     | Small       |
| singletask_extra3 vs uniform_extra3                                   | 0.0093          | 0.1905  |              | No                     | 0.704     | Medium      |
| singletask_extra3 vs no_rl_full_train                                 | 0.0484          | 0.0386  | *            | No                     | 1.357     | Large       |
| singletask_extra3 vs original_nolnitWeights                           | 0.0421          | 0.0096  | **           | No                     | 2.082     | Large       |
| singletask_extra3 vs original                                         | 0.0365          | 0.0271  | *            | No                     | 1.523     | Large       |
| singletask_extra3 vs vivit                                            | 0.1381          | 0.0003  | ***          | Yes                    | 5.413     | Large       |
| singletask_extra3 vs attention_pool_extra3_full_train                 | -0.0082         | 0.3867  |              | No                     | -0.434    | Small       |
| singletask_extra3 vs attention_pool_extra4_k1                         | 0.0115          | 0.3353  |              | No                     | 0.489     | Small       |
| singletask_extra3 vs attention_pool_extra4_k8                         | 0.0148          | 0.2376  |              | No                     | 0.620     | Medium      |
| uniform_extra3 vs no_rl_full_train                                    | 0.0391          | 0.0656  |              | No                     | 1.125     | Large       |
| uniform_extra3 vs original_nolnitWeights                              | 0.0329          | 0.0101  | *            | No                     | 2.052     | Large       |
| uniform_extra3 vs original                                            | 0.0272          | 0.0317  | *            | No                     | 1.448     | Large       |
| uniform_extra3 vs vivit                                               | 0.1289          | 0.0001  | ***          | Yes                    | 6.652     | Large       |
| uniform_extra3 vs attention_pool_extra3_full_train                    | -0.0175         | 0.0062  | **           | No                     | -2.360    | Large       |
| uniform_extra3 vs attention_pool_extra4_k1                            | 0.0022          | 0.8539  |              | No                     | 0.088     | Negligible  |
| uniform_extra3 vs attention_pool_extra4_k8                            | 0.0055          | 0.5901  |              | No                     | 0.262     | Small       |
| no_rl_full_train vs original_nolnitWeights                            | -0.0063         | 0.7703  |              | No                     | -0.140    | Negligible  |
| no_rl_full_train vs original                                          | -0.0119         | 0.2295  |              | No                     | -0.634    | Medium      |
| no_rl_full_train vs vivit                                             | 0.0897          | 0.0032  | **           | No                     | 2.833     | Large       |
| no_rl_full_train vs attention_pool_extra3_full_train                  | -0.0566         | 0.0273  | *            | No                     | -1.519    | Large       |
| no_rl_full_train vs attention_pool_extra4_k1                          | -0.0369         | 0.1382  |              | No                     | -0.827    | Large       |
| no_rl_full_train vs attention_pool_extra4_k8                          | -0.0336         | 0.1752  |              | No                     | -0.736    | Medium      |
| original_nolnitWeights vs original                                    | -0.0056         | 0.6918  |              | No                     | -0.191    | Negligible  |
| original_nolnitWeights vs vivit                                       | 0.0960          | 0.0003  | ***          | Yes                    | 5.067     | Large       |
| original_nolnitWeights vs attention_pool_extra3_full_train            | -0.0503         | 0.0010  | **           | No                     | -3.837    | Large       |
| original_nolnitWeights vs attention_pool_extra4_k1                    | -0.0306         | 0.1369  |              | No                     | -0.830    | Large       |
| original_nolnitWeights vs attention_pool_extra4_k8                    | -0.0273         | 0.1403  |              | No                     | -0.821    | Large       |
| original vs vivit                                                     | 0.1016          | 0.0002  | ***          | Yes                    | 6.083     | Large       |
| original vs attention_pool_extra3_full_train                          | -0.0447         | 0.0072  | **           | No                     | -2.265    | Large       |
| original vs attention_pool_extra4_k1                                  | -0.0250         | 0.1996  |              | No                     | -0.686    | Medium      |
| original vs attention_pool_extra4_k8                                  | -0.0217         | 0.2284  |              | No                     | -0.635    | Medium      |
| vivit vs attention_pool_extra3_full_train                             | -0.1463         | 0.0000  | ***          | Yes                    | -9.551    | Large       |
| vivit vs attention_pool_extra4_k1                                     | -0.1267         | 0.0029  | **           | No                     | -2.901    | Large       |
| vivit vs attention_pool_extra4_k8                                     | -0.1233         | 0.0024  | **           | No                     | -3.065    | Large       |
| attention_pool_extra3_full_train2 vs attention_pool_extra3_full_train | 0.0197          | 0.2400  |              | No                     | 0.617     | Medium      |
| attention_pool_extra3_full_train2 vs attention_pool_extra3_full_train | 0.0230          | 0.1265  |              | No                     | 0.861     | Large       |
| attention_pool_extra4_k1 vs attention_pool_extra4_k8                  | 0.0033          | 0.4847  |              | No                     | 0.344     | Small       |